



Classifying retinal diseases via pyramid vision graph convolutional network for optical coherence tomography images

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Abstract: Recent advancements have seen a significant focus on using deep neural networks for classifying retinal diseases in optical coherence tomography (OCT) images. However, traditional deep neural networks treat images as grid or sequential structures, limiting their flexibility in capturing irregular and complex objects, resulting in suboptimal performance in practical applications. To address this issue, we propose a novel visual neural network model with a pyramid structure, called pyramid vision graph convolutional networks (PVGCN). This model enhances the correlations between structures by segmenting images into multiple nodes and connecting the nearest nodes. Specifically, it consists of two core components: 1) vision graph block and 2) pyramid structure. The vision graph block, composed of a grapher block and a feed-forward network (FFN), uses graph convolution methods to divide the image into multiple regions, treating them as nodes and representing the image as graph data. The graph constructed based on nodes can capture relationships between nodes without positional restrictions, better representing the irregular structure of retinal tissue. The FFN module improves the over-smoothing phenomenon in the grapher stage, enabling more accurate classification. The pyramid structure decomposes OCT images into a series of sub-images at different scales, integrating features at different scales to obtain a comprehensive feature representation of retinal hierarchical structure information. This structure can replace the extraction of higher-dimensional features in a large model by integrating features at different scales, significantly reducing the number of parameters. We conducted extensive experiments on two different datasets. The experimental results show that the proposed PVGCN achieved accuracies of 0.9954 and 0.9787 on the two datasets, respectively, surpassing existing methods. Additionally, the model demonstrated recognition capabilities comparable to those of human experts in the experiments, effectively identifying retinal diseases in OCT images.

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1. Introduction

Retinal diseases such as Age-related Macular Degeneration (AMD), Choroidal Neovascularization (CNV), Drusen, and Diabetic Macular Edema (DME) can severely impair vision if not promptly treated, significantly affecting patients' lives [1]. As demographic trends shift towards longer life expectancies and greater elderly populations, the incidence of AMD and DME is expected to increase. Optical Coherence Tomography (OCT), with its high resolution, non-contact, and non-destructive characteristics, utilizes the principle of interference from low-coherence light sources to capture detailed images of the retina, playing an indispensable role in diagnosing these conditions [2–5]. OCT enables precise visualization of retinal layers and helps in differentiating the morphology and thickness of crucial components such as the nerve fiber layer, ganglion cell layer, and others, providing invaluable support in the diagnosis and treatment of retinal diseases [6,7]. Despite the significant advantages of OCT technology in diagnosing retinal

diseases, the generation of approximately 30 million OCT images worldwide each year presents a substantial challenge in terms of image analysis [8]. This vast amount of data places immense pressure on the identification process, particularly in regions suffering from a shortage of ophthalmologists, rendering manual examination by doctors nearly impracticable [9]. Such circumstances underscore the urgent need for scalable solutions to manage and interpret these images efficiently.

In recent years, deep learning has made significant advancements in fields such as natural language processing, speech recognition, and computer vision [10]. These methods have also been used in the automatic classification of retinal diseases in OCT images, revolutionizing ophthalmic diagnostics. Wang et al. applied techniques such as CliqueNet, DPN, DenseNet, ResNet, and ResNeXt on OCT datasets, using 80% of the data for training [11]. Meanwhile, Hassan et al. enhanced the accuracy of retinal disease classification using a Bayesian multi-objective deep learning approach, achieving remarkable accuracies and F1 scores of 0.9826 and 0.9846 across six datasets [12]. Sunija et al. utilized a deep convolutional network with six blocks to achieve a 99.69% accuracy rate on a dataset of 83,484 OCT images [13]. Furthermore, Khan et al. introduced a combination of feature selection and deep learning to accurately classify five different retinal diseases with over 99.1% accuracy in their dataset [14]. Moreover, Li et al. [15] employed graph-based deep learning methods for graph representation learning in biomedical research, while Lee et al. [16] utilized graph deep learning to derive prognostic contextual histopathological features from whole-slide tumor images. Existing deep learning approaches for classifying retinal lesions in OCT images still have room for improvements. They generally treat OCT images with isotropic structures based on a fixed scale, viewing all image parts as equally important without distinguishing information at different scales [17]. This leads to simple convolutional operations that do not account for feature variations across scales, potentially missing crucial detail information and resulting in inaccurate outcomes. To address these limitations, there is a need for a multi-scale approach that captures more hierarchical information. Additionally, existing deep learning models perceive images as grid or sequential structures, which restricts their ability to capture irregular and complex objects, reducing flexibility. This may result in satisfactory performance on simple datasets but poor results on complex ones. Moreover, while methods like VGG-16 achieve good classification results, their large parameter counts lead to substantial computational resource consumption which may not be available for less-developed regions [13]. Thus, there is a necessity for more lightweight models that reduce computational demands without compromising performance.

To reduce computational resource consumption, improve utilization, and handle multiple complex OCT datasets, we propose a lightweight visual neural network model with a pyramid structure (PVGCN) for automatic classification of OCT cases, inspired by the ViG model. Our proposed PVGCN is based on Graph Convolutional Networks (GCN), which can be seen as a special case of grid and sequence structures. Traditional deep learning methods for processing OCT images, such as the commonly used GNN, can only order pixels based on spatial position. In PVGCN, nodes can connect freely without being constrained by local spatial structures. This flexibility allows the model to connect different structures in the retina, such as the optic cup, optic disc, and blood vessels, forming a graph structure that effectively integrates positional information. However, conventional neighbor-based GCNs continuously aggregate features from neighboring nodes in each iteration, leading to feature information gradually spreading and becoming homogeneous. This may result in the loss of micro-differences in the original graph structure when learning node features, making it difficult to distinguish information between nodes and potentially causing over-smoothing, thus leading to poor performance. To address this, our PVGCN method incorporates nonlinear activation and feature transformation within the Vision Graph block to improve GCN, alleviating layer collapse and over-smoothing phenomena [18]. The graph convolution layer effectively aggregates feature information from neighboring

nodes, while the feed-forward network (FFN) transforms and processes node features, enhancing feature diversity and information exchange capabilities.

Moreover, commonly used structures such as isotropic structures are fixed-scale image processing methods that do not distinguish between different scales when processing OCT images [19]. They only perform simple convolution operations on images without considering the differences in features at various scales, which may lead to a lack of detailed and flexible information capture within the image. To effectively capture multi-scale features of images, the PVGCN model employs a pyramid architecture with progressive reduction [20]. This architecture decomposes OCT images into multiple sub-images at different scales, then processes each scale separately. This approach captures feature information across different scales more comprehensively, linking positional information of retinal tissues to improve the accuracy and reliability of image processing. Compared to isotropic structures, the pyramid structure can better handle fine features in images, improving the quality and accuracy of OCT images. This architecture enables PVGCN to better identify various structures within the retina when processing OCT images, thereby achieving superior learning outcomes.

Additionally, in the Vision Graph block of our proposed PVGCN, the image is divided into many blocks and treated as nodes, converting the image into graph data. Unlike traditional CNNs, which treat each pixel of the image as a node, this approach significantly reduces the number of parameters in PVGCN. The commonly used CNN-5 has 138 million parameters, and such a large number of parameters require high computational resources. The PVGCN has 10.7 million parameters, using only 7% of the parameters in VGG-16. This significantly saves computational resources without affecting the final performance. In particular, we represent images using a graph structure and directly use Graph Convolutional Networks to process image data, gradually extracting features and achieving object recognition and detection. This image processing workflow can better capture information and relationships within images, enhancing the accuracy and efficiency of visual tasks. Finally, we conducted extensive experiments on two public datasets, achieving the highest accuracy in both cases. The experiments validated that this architecture outperformed other comparative deep learning methods, reaching the level of human experts.

The main contributions of this paper can be summarized as follows:

- We propose the Vision Graph block for processing complex OCT images. This block, consisting of a Grapher block and an FFN, uses graph convolution techniques and nonlinear transformations to effectively aggregate and update node information, alleviating layer collapse and over-smoothing. This design allows PVGCN to flexibly capture irregular and complex structures, demonstrating superior performance in handling complex OCT images.
- We propose a pyramid architecture with progressive reduction in PVGCN, which decomposes OCT images into multi-scale sub-images for processing. Compared to methods that consider only a single scale, this multi-scale approach allows PVGCN to better understand and capture complex information, enhancing the global view and leading to superior learning outcomes.
- The proposed PVGCN uses a pyramid structure to fuse features extracted at different levels, resulting in richer feature representations and reducing the number of parameters by avoiding the need for a large model to extract higher-dimensional features. Thus, it is more lightweight than existing approaches.

2. Methodology

As shown in Fig. 1, we demonstrate the overview of PVGCN for retinal disease classification in OCT images. Taking AMD, DME, and NORMAL as examples, we first segment images from three different types of OCT datasets into nodes in the Image Instantiation step, and then connect these nodes to form a graph structure. For constructing the graph structure, we divide an OCT image of size $H \times W \times 3$ into N blocks and convert them into feature vectors $\mathbf{x}_i \in \mathbb{R}^D$, denoted as $X = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N]$, where D represents the feature dimension, and $i = 1, 2, \dots, N$. Finally, we connect the nodes to obtain the Graph, as shown in Step 1.

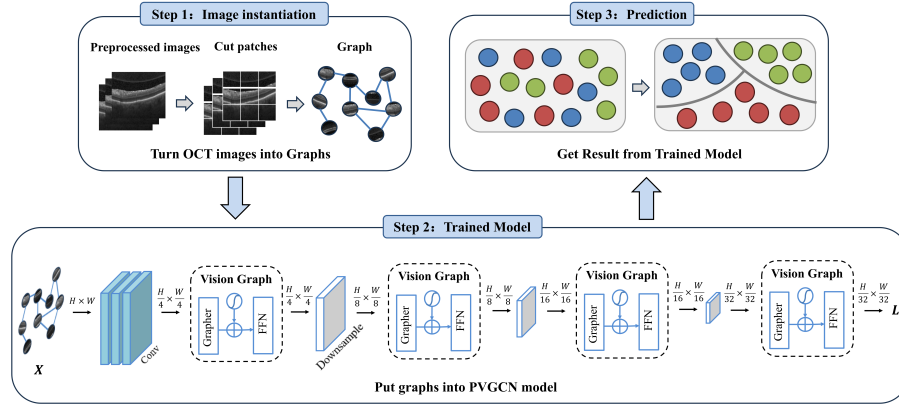


Fig. 1. The following is a synopsis of the method. Step I: The OCT image is first divided into patches, which are then structured into a graph. Step II: The graph data is processed by the PVGCN, where graph information and node features are aggregated and updated. Simultaneously, scale reduction is applied, forming a pyramid structure. Step III: Model performance is evaluated using predictions on a designated test set.

In Step 2, through the above steps, the image is output as graphical data X . We input X into the PVGCN containing multiple Vision Graph modules. The Vision Graph blocks consist of two main components: the Grapher module and the FFN module. The Grapher module uses graph convolution to aggregate and update graph information. The FFN module includes two linear layers for node feature transformations. Together, these components process and refine the graph data. In the feature extraction process, a Downsample block is added after each FFN module to achieve scale reduction. As the number of layers increases, the extraction space size is progressively reduced, ultimately forming a progressively shrinking pyramid structure. The model is trained based on the loss value L during this training process.

In Step 3, we input the test set data into the trained model for prediction, completing the retinal disease classification task. We will detail the specific components of PVGCN in Section 2.1.

2.1. PVGCN

In our proposed PVGCN method, Vision Graph blocks are used as key components, primarily achieving graph-level feature extraction. They are mainly composed of stacked Grapher and FFN modules, and for the output results, a spatial reduction strategy is used to construct Downsample blocks to handle a large number of nodes. Additionally, we adopted a progressively shrinking pyramid structure, which is more suitable for dense tasks and reduces computational load. Below, we will introduce the implementation of graph structure processing, Vision Graph block structure, Downsample block, and pyramid structure.

2.1.1. Graph structure processing

We start with features to construct a graph structure $G=(\mathcal{V}, \mathcal{E})$ based on feature representation. The graph structure is constructed by dividing the image into N patches. We treat these features as a set of unordered nodes, denoted as $\mathcal{V} = \{v_1, v_2, \dots, v_N\}$. In terms of node selection, the K-Nearest Neighbors (KNN) algorithm is typically employed to select the K neighboring nodes for each node. For each node v_i , we find its K nearest nodes $\mathcal{N}(v_i)$ and add an edge e_{ji} from v_j to v_i for all $v_j \in \mathcal{N}(v_i)$. Finally, we obtain a graph $G = (\mathcal{V}, \mathcal{E})$, where $\mathcal{E}$ represents all the edges. Edges are created in the graph to connect the selected nodes, thereby linking the nodes. The weights of the edges can be determined based on the similarity or distance between node features. In this way, the image is represented as a graph structure, which can then be utilized to extract its representation.

Since normal objects are not regular shapes, using a graph structure to handle complex objects can be more flexible. Meanwhile, viewing an object as a combination of multiple parts, the graph structure can effectively establish the relationships between them.

2.1.2. Vision graph block structure

Based on the aforementioned graph structure processing, we input the graphical data into PVGCN, where the Vision Graph blocks process the input. Below, we detail the roles of each part of the Vision Graph block.

Grapher Module The input OCT image is first downsampled by the Stem module to generate a feature map. Each spatial location (pixel) in the feature map is considered as a node in the map, and the node features are represented by the multichannel feature vector at that location.

The process begins by selecting K neighboring nodes using the K-Nearest Neighbors (KNN) algorithm. Neighborhood sampling using mixed-radius (MR) strategy to balance local and global connectivity. Subsequently, the features of the neighboring nodes are extracted, and a weighted summation is performed on these features to obtain the aggregated representation.

Using aggregation, features between neighboring nodes are exchanged to ultimately output the predicted Y . The specific operations are as follows:

$$G' = F(G, W) = \text{Update}(\text{Aggregate}(G, W_a), W_b) \quad (1)$$

where W_a and W_b represent the weights of the aggregation and update operations, respectively. These can be viewed as a graph convolution layer and a fully connected layer with hidden bias terms. This step allows for the aggregation of features between neighboring nodes, then the calculation and updating of the nodes. Meanwhile, using the underlying mapping H , the result transformed by F is taken as the residual result, and the output result is added pointwise as new input.

$$G = H(G', W') = F(G', W') + G' = G'^{\text{res}} + G' \quad (2)$$

$$N^{(d)}(x_i) = x_1, x_{1+d}, x_{1+2d}, \dots, x_{1+(k-1)d} \quad (3)$$

where $N^{(d)}(x_i)$ represents the set of x_i . Here, we use the max relative graph convolution with simplicity and efficiency [21], which can efficiently handle this part. K and d denote the $k \times d$ nearest neighbors in the first order

$$xi' = \left(xi, \text{Max} \left(xi, N^{(d)}(xi), W_a \right) \right) W_b \quad (4)$$

$$\text{Max}(\cdot) = \left[x_i, \max \left(x_j - x_i | j \in N^{(d)}(x_i) \right) \right] \quad (5)$$

Through the above processing, the xi' can be represented as $\text{Gr}(X)$. Specifically, for $X \in \mathbb{R}^{N \times D}$, the new module is described by the following formula:

$$H = \sigma(\text{Gr}(XW_1)) W_2 + X \quad (6)$$

where σ represents the activation function, and common activation functions include ReLU and Sigmoid. W_1 and W_2 represent the weights of the fully connected layers.

FFN Module Many GCN models repeatedly use multiple graph convolution layers to extract features for aggregation operations. However, through continuous iterative updates, GCN causes the features of neighboring nodes to continuously aggregate, making feature information gradually similar, reducing the differences between nodes, and ultimately leading to the over-smoothing phenomenon. This reduces the prominence of feature nodes and results in performance degradation [18]. To further enhance feature transformation capabilities and alleviate the over-smoothing phenomenon, we leverage FNN on each node.

The FFN is a multi-layer perceptron with two fully connected layers. It captures complex relationships within sequences and performs fully connected transformations on encoded and decoded positional data. This enhances positional representation and feature extraction. The FFN processes input features and maps them to new representations while preserving residual connection information. This helps the network learn features more effectively. These steps form the module's forward propagation process. This approach also helps reduce the over-smoothing phenomenon.

$$D = \sigma(HW_1)W_2 + H \quad (7)$$

where W_1 and W_2 represent the weights of the fully connected layers. The above formulas combine to form the Vision Graph block. By cyclically stacking different numbers of Grapher blocks and FFN blocks, the Vision Graph block is formed. Each fully connected layer and graph convolution layer is followed by batch normalization. Combining the characteristics of both, maintaining feature diversity, helps in the deep learning process.

subsubsectionDownsample Block In practical computations, the data volume can become excessively large. If the output scale is not reduced, it can easily lead to a waste of computational resources. To reduce space and save computational resources, we used a spatial reduction strategy to decrease the generated space. This operation helps reduce the parameter size, making the model more lightweight [20]. In the specific construction of the model, to reduce the scale of the output feature map, we designed a downsample block to reduce the dimensionality of the output. This block ultimately halves the scale of the feature map. Additionally, batch normalization is applied to each layer. This process normalizes the layers, preventing overfitting while enhancing the model's training speed and performance.

2.1.3. Pyramid structure

Classic Transformers have demonstrated significant success in natural language processing. However, in object detection, current popular detectors face different challenges. These detectors often require multi-scale and high-resolution images to achieve effective target detection. The pixel resolution in images is much higher than the word resolution in text. Due to the output of single-scale and low-resolution feature maps and high computational resource consumption, Transformers have significant limitations in handling image tasks [22]. Existing experiments have shown that pyramid structures perform well in handling visual image problems [23].

To overcome these limitations, we propose a progressively decreasing pyramid structure. Similar to traditional CNN backbone networks, our proposed pyramid architecture generates different scale maps in four stages, sharing a similar structure [24]. We introduced relative position encoding, which better expresses the positional relationships between nodes [25].

For an image of size $H \times W \times 3$, in the first stage, we divide it into $\frac{HW}{4^2}$ patches of size $4 \times 4 \times 3$, and then linearly project these patches to obtain embedding blocks of size $\frac{HW}{4^2} \times C_1$. Finally, the embedding blocks obtained in the previous step are processed together with position embeddings to reform a feature F of size $\frac{H}{4} \times \frac{W}{4} \times C_1$. For the subsequent stages, the output result of the previous stage is used as input for the same operations. Our proposed pyramid structure builds on these operations. As the layers deepen, it extracts features with progressively smaller spatial

dimensions. This design effectively incorporates the multi-scale characteristics of the image. Therefore, our proposed PVGCN model adopts a pyramid structure. This structure can acquire high-scale and high-resolution features, providing unique advantages for complex OCT image recognition problems.

3. Evaluation

3.1. Data description

In this paper, to validate the performance of our method, we chose to use two popular retinal disease detection datasets for evaluation. These two datasets have their own characteristics, and by comparing the results on these datasets, we can more effectively analyze the outcomes. Some sample OCT images of the two datasets are shown in Fig. 2.

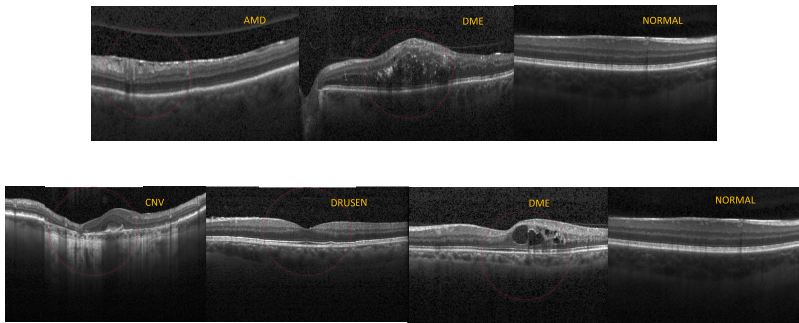


Fig. 2. Types of retinal cases in the dataset.

The first dataset (BOE dataset) was collected by Srinivasan et al. [26] from 45 subjects invited from Duke University, Harvard University, and the University of Michigan, collecting OCT images. This dataset contains 723 AMD images, 1101 DME images, and 1407 NORMAL images, covering OCT images from 15 normal patients, 15 AMD patients, and 15 DME patients. These images were obtained through different scan numbers and scan locations. Since the BOE dataset is a three-class classification task, we set the number of outputs in the softmax layer to 3 in our experimental setup.

The second dataset (CELL dataset) comes from kermany [27]. A total of 84484 OCT images were collected from 5319 subjects, including 37455 CNV images, 11498 DME images, 8866 DRUSEN images, and 26565 NORMAL images. The CELL dataset is a four-class classification task, containing four types of retinal diseases, so we set C to 4.

Specifically, in this experiment, we divided both datasets into 80% training set, 10% test set, and 10% validation set. For data processing, the input image size is $224 \times 224 \times 3$. Since the original sizes of the OCT images are different, the datasets need to be preprocessed for subsequent training. First, the OCT image sizes are standardized, and then all images are randomly shuffled. Data selection is done randomly to ensure that the same image does not appear in other categories, thereby avoiding any impact on the experimental results.

3.2. Experimental setup

In this paper, we performed fully supervised retinal disease classification tasks on two datasets. To evaluate and compare the performance of our method, we selected several state-of-the-art deep learning methods, including VGG-16 [28], CNN-5 [29], AlexNet [28], MobileNetV2 [30], Graph Transformer [31], and ResNet-18 [32].

VGG-16 is a classic block-based model, with 13 convolutional layers and 5 pooling layers responsible for feature extraction, and the final 3 fully connected layers responsible for completing

the classification task. CNN-5 is a standard convolutional network used for handwritten digit recognition, consisting of two convolutional layers and three fully connected layers. AlexNet was the first to increase network depth and incorporate the use of activation functions, adding a normalization operation between convolutional and pooling layers, and employing data augmentation and Dropout. ResNet-18 adds extra connections to solve issues of gradient vanishing and exploding in deep neural network training, with its basic framework being an 18-layer deep ResNet network. MobileNetV2 further enhances the efficiency and performance of the model by introducing innovations such as Inverted Residuals and Linear Bottleneck layers. Graph Transformer is a model that applies the Transformer architecture to graph data. It combines the features of Transformer's Self-Attention and Graph Neural Networks to efficiently process nodes, edges, and global information in graph-structured data. Using these seven classic deep learning methods as benchmarks, we can effectively analyze and evaluate the performance of our proposed PVGCN. In this paper, we use accuracy, precision, recall, and F1 score as evaluation metrics.

3.3. Implementation details

This model trains on the BOE and CELL datasets, with a batch size of 45 for labeled data, a learning rate of 0.01, momentum of 0.9, and 200 training epochs. In order to improve the stability of the model, we adopt some data enhancement strategies, i.e., level-flipping, color dithering, and stochastic erasure. Specifically, the probability of horizontal flip is set to 0.5, and the color dithering factor is set to 0.4. The random erasure *lai* is used to help the model generalize better and to enhance its robustness to partially occluded or missing information. During training, using the BOE dataset as an example, the dataset is trained with a batch size of 45, randomly sampling the dataset until all data is used. For the model, we use ReLU and Softmax as activation functions and also use Exponential Moving Average (EMA) to track the moving averages of model parameters, continuously updating during training to improve the model's generalization performance. Additionally, Dropout and batch normalization operations are used to prevent overfitting. The code is written in Python 3 and runs on an NVIDIA GeForce RTX 4070 Ti GPU.

3.4. Comparison with benchmark methods

To analyze our proposed PVGCN method from multiple aspects, we present the confusion matrices for processing the two datasets in Fig. 3. From the figures, it can be observed that in the BOE dataset, only the third class has an error of 0.07. In the CELL dataset, the classification performance for the four categories reached 0.98, 0.93, 0.83, and 1, respectively.

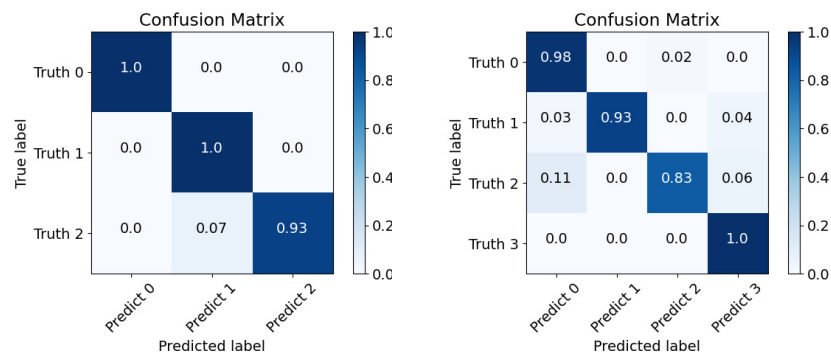


Fig. 3. The numerical value in each column represents the proportion of the number of real data that is predicted to be that class. The left picture shows the confusion matrix of PVGCN in the BOE dataset, and the right picture shows the confusion matrix of PVGCN in the CELL dataset.

The experimental results compared with the benchmark methods are shown in Tables 1 and 2. It can be observed that the performance of PVGCN on these two datasets is superior to all other benchmark methods. This demonstrates the effectiveness of PVGCN in detecting retinal diseases using OCT images.

Table 1. The effects of different algorithms on the BOE data set. Bold indicates the best

Method	ACCURACY	PRECISION	RECALL	F1
ResNet-18 [32]	0.8889	0.8508	0.8133	0.8258
AlexNet [28]	0.4729	0.1465	0.3431	0.1947
VGG-16 [33]	0.6757	0.6224	0.5425	0.5409
CNN-5 [29]	0.9595	0.9786	0.9590	0.9676
MobileNetV2 [30]	0.9054	0.9127	0.8889	0.8919
Graph Transformer [31]	0.9865	0.9869	0.9865	0.9864
PVGCN	0.9954	0.9918	0.9877	0.9896

Table 2. The effects of different algorithms on the CELL data set. Bold indicates the best

Method	ACCURACY	PRECISION	RECALL	F1
ResNet-18 [32]	0.9694	0.9431	0.9296	0.9355
AlexNet [28]	0.8883	0.8379	0.6428	0.6397
VGG-16 [33]	0.9735	0.9595	0.9393	0.9488
CNN-5 [29]	0.9570	0.9272	0.9315	0.9291
MobileNetV2 [30]	0.9658	0.9494	0.9291	0.9389
Graph Transformer [31]	0.9592	0.9591	0.9592	0.9586
PVGCN	0.9787	0.9650	0.9555	0.9602

Additionally, our proposed PVGCN method achieved accuracies of 0.9954 and 0.9787, precision of 0.9918 and 0.9650, recall of 0.9877 and 0.9555, and F1 scores of 0.9896 and 0.9602 on the two datasets, significantly outperforming all benchmark methods.

In both datasets, the accuracy of the PVGCN model is higher than that of other methods, demonstrating superior performance. This result indicates that the PVGCN model has significant effectiveness in handling OCT image recognition tasks. Since the CELL dataset contains 84484 OCT images from 5319 subjects, the large data volume may lead to some decline in data results. However, the CELL dataset is a time-consuming and labor-intensive dataset, gathering data from major laboratories worldwide, making it difficult to acquire again. In practical applications, the BOE dataset is more commonly used for training.

3.5. Analysis of model performance with different sample ratios

As shown in Table 3, to visually display the differences of several methods on the CELL dataset, we tested these methods using 0.8, 0.5, 0.2, 0.1, and 0.05 of the original dataset volume, observing the performance differences of the methods under different data volumes. This analysis can help us understand the impact of different data volumes on model performance, thereby better evaluating the robustness and generalization ability of the model.

From the data results, it can be observed that as the data volume gradually decreases, the differences between the various methods become more apparent. When the data volume is 0.05 (approximately 4224 OCT images), the difference in accuracy between the algorithms is the greatest. From the final results, it can be seen that PVGCN outperforms VGG-16 by nearly

Table 3. Accuracy results of different algorithms on the CELL data set with different sample ratios. Bold indicates the best

Method	0.8	0.5	0.2	0.1	0.05
ResNet-18 [32]	0.9689	0.9626	0.9591	0.9414	0.9308
AlexNet [28]	0.8935	0.8640	0.8595	0.8535	0.8510
VGG-16 [33]	0.9710	0.9605	0.9500	0.9591	0.9312
CNN-5 [29]	0.9645	0.9578	0.9477	0.9299	0.9320
MobileNetV2 [30]	0.9457	0.9471	0.9186	0.8082	0.7564
Graph Transformer [31]	0.9475	0.9591	0.9536	0.9655	0.9455
PVGCN	0.9772	0.9727	0.9690	0.9663	0.9599

three percentage points. Therefore, we can conclude that PVGCN performs better than the other methods on the BOE and CELL datasets.

In the experiment, we observed that on the smaller BOE dataset, our method achieved an accuracy of 0.9954, which is about two percentage points higher than the other methods. However, on the larger CELL dataset, this difference becomes smaller. It is worth mentioning that constructing the CELL dataset requires a significant amount of effort and time, making it difficult to replicate. Our experiments demonstrate that as the dataset size decreases, the accuracy advantage of our method gradually becomes apparent.

In practical applications, most datasets are similar in size to the BOE dataset, so our method still holds a significant advantage. Additionally, compared to other high-accuracy methods, our model has fewer parameters and requires less computational resources, making it more suitable for practical applications. While ensuring high precision, our method can achieve the desired results in a shorter time.

3.6. Model size comparison

When processing different medical OCT images, given varying data volumes and computational resources, we usually aim to achieve high-precision results in less time and resources. We have conducted additional experiments comparing PVGCN with the second-best performing models on each dataset in terms of model parameters and inference speed.

On the BOE dataset that contains 223 testing samples, we compared PVGCN (11.47 million parameters) with Graph Transformer (13.81 million parameters) that is the second best model. The average inference time for PVGCN was 0.53 seconds, significantly faster than the 0.89 seconds required by Graph Transformer. This demonstrates that PVGCN offers both parameter efficiency and real-time inference advantages.

On the CELL dataset that contains 7783 samples, we compared PVGCN with VGG-16 (the second best model), which has significantly more parameters (138.36 million). Despite this, the inference time difference is modest—12.47 seconds for PVGCN versus 12.76 seconds for VGG-16. This can be attributed to the graph construction overhead in PVGCN. However, PVGCN still maintains a lower inference time, while requiring only 8% of VGG-16's parameter count, leading to significantly lower memory consumption and better compatibility with resource-limited environments.

3.7. Comparison with human experts

In real life, the classification of retinal diseases typically requires experienced human experts. In the CELL dataset, Kermany et al. invited six experienced human experts from academic ophthalmology centers to classify 1000 OCT images [34]. By comparing with these human

experts, we can evaluate the performance of the proposed method in recognizing different retinal diseases through OCT images.

According to the information you provided, as shown in Fig. 4, the proposed method performed excellently in actual testing. The best test accuracy was approximately 0.997, and the lowest accuracy was around 0.921. The method's accuracy ranked among the top three, reaching the level of human experts, indicating that the method has high accuracy and reliability in retinal disease recognition tasks and can be effectively applied in practical use.

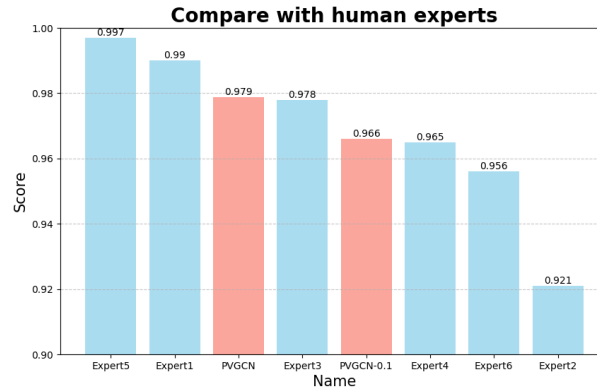


Fig. 4. Histogram comparing the accuracy of PVGCN and six famous human experts in the CELL data set

These results show that the proposed method has high performance in retinal disease recognition and can serve as an effective tool to assist doctors in diagnosis and classification, bringing significant progress to the field of medical image analysis.

3.8. Ablation study

The paper proposes that PVGCN consists of two core components: Vision Graph block and pyramid structure, which play crucial roles. We verified the advantages of these two core components through ablation experiments.

In our proposed method, we enhance the feature extraction by adding more feature transformations to the GCN module, resulting in an upgraded module named the Vision Graph block. These components play a significant role throughout the entire method. Through ablation experiments as shown in Table 4, when utilizing only the Vision Graph block without the pyramid structure, we achieved an accuracy of 0.9909 compared to the 0.9774 accuracy of the non-upgraded GCN, demonstrating performance improvement. This indicates the important role of the Vision Graph in PVGCN, enhancing feature transformations to better adapt to the task of retinal lesion recognition.

Table 4. The effects of modules in PVGCN

Method	Accuracy
GCN	0.9774
Vision Graph	0.9909
GCN + Pyramid	0.9861
Vision Graph + Pyramid (Proposed)	0.9954

Furthermore, we validate the advantages of the pyramid structure for PVGCN. The pyramid structure gradually reduces computational complexity and captures features from multiple

scales to enhance accuracy. By incorporating the pyramid structure on top of both GCN and Vision Graph separately, we compared the performance with and without the pyramid structure. Experimental results demonstrate that the accuracy using the Vision Graph under the pyramid structure is 0.9954, compared to 0.9909 without the pyramid structure, indicating performance improvement with the pyramid structure. The data from the table illustrates that both the Vision Graph and the pyramid structure play crucial roles in improving performance.

To further investigate the contribution of each scale in our multi-scale fusion design, we conducted an additional experiment to evaluate the performance of our model when using different combinations of scales. As shown in Table 5, we tested four settings: (1) only the 1st scale, (2) 1st and 2nd scales, (3) 1st to 3rd scales, and (4) all four scales. The results demonstrate a clear trend of performance improvement with the addition of each scale. Specifically, the accuracy increases from 0.5638 (1st scale only) to 0.9787 (all four scales), and similar improvements are observed across precision, recall, and F1 score metrics.

Table 5. Experimental results with different scales for multi-scale fusion

Method	ACCURACY	PRECISION	RECALL	F1
1st scale only	0.5638	0.5688	0.4465	0.4655
1st and 2nd scales	0.6230	0.7872	0.4625	0.5021
1st, 2nd and 3rd scales	0.9047	0.9110	0.8450	0.8687
All four scales	0.9787	0.9650	0.9555	0.9602

This progressive enhancement confirms that each additional scale introduces complementary feature representations, effectively capturing both local details and global contextual information. The substantial leap in performance between using two scales (accuracy = 0.6230) and three scales (accuracy = 0.9047) underscores the importance of mid- to high-level features, while the final boost from adding the fourth scale suggests that deeper semantic information further refines the classification decision. Note that the choice of using four scales is experimentally justified, striking a balance between performance gain and computational efficiency.

3.9. Sensitivity analysis

The number of neighbors, K , is a hyperparameter that controls the aggregation range when constructing the graph. It represents the number of adjacent nodes. Adjusting the value of K can affect the degree of information exchange. Too few neighbors can reduce information transfer, while too many neighbors may lead to over-smoothing. Therefore, we adjusted the number of K and conducted a comparative analysis of the impact of different K values on accuracy.

In the experiment, we adjusted K from 3 to 12 and presented the results in Table 6. The experimental results show that when the number of neighbor nodes is between 9 and 12, the classification task is performed well, with the best results when K is set to 9. These experimental results further validate the impact of selecting the number of neighbors, K , on model performance in the PVGCN method, providing useful guidance for optimizing model design.

Table 6. Impact on accuracy under different k values

K	3	6	9	12
Accuracy	0.9901	0.9864	0.9954	0.9954

To assess the impact of class imbalance in the CELL dataset, we conducted additional experiments using a class-balanced training set. This was achieved via over-sampling of the minority classes using data augmentation techniques such as rotating, brightness adjustment, and

random cropping. After applying this strategy, the training data of each class was brought to a similar scale.

The revised performance metrics on the CELL dataset are shown in Table 7. Compared with the original results (Accuracy: 0.9787, Precision: 0.9650, Recall: 0.9555, F1 Score: 0.9602), the balanced training achieved improved Recall (0.9713) and F1 Score (0.9653), with slight increases in Accuracy (0.9802) and stable Precision (0.9598). These results suggest that class balancing enhances the model's ability to detect minority class instances without compromising overall performance.

Table 7. Effect of data augmentation-based class balancing on PVGCN performance for the CELL dataset

Method	ACCURACY	PRECISION	RECALL	F1
Original PVGCN	0.9787	0.9650	0.9555	0.9602
PVGCN with balanced training data	0.9802	0.9598	0.9713	0.9653

4. Conclusion

In this paper, we proposed a novel pyramid-structured visual neural network model (PVGCN) for retinal disease recognition. Using the Vision Graph block, which combines the Grapher module and FFN module as basic units, the model employs graph convolution techniques to aggregate and update graph information and incorporates nonlinear transformations to address the over-smoothing issue in images. The progressively shrinking pyramid structure of PVGCN better leverages the characteristics of the graph structure, naturally capturing the connections between various structures from multiple scales. Additionally, by employing spatial reduction, the consumption of computational resources is reduced, making PVGCN more lightweight. On two popular OCT datasets (BOE and CELL), PVGCN achieved the best results of 0.9954 and 0.9787, respectively, surpassing other compared deep learning methods. Moreover, on the CELL dataset, the PVGCN network reached the level of top human experts. Additionally, through ablation experiments, we validated the roles of the utilized modules in the model.

While the proposed PVGCN has demonstrated exceptional performance in classifying retinal diseases from OCT images, it is important to acknowledge its limitations. A key challenge is the model's ability to generalize to other medical imaging tasks, such as Magnetic Resonance Imaging (MRI) or Computed Tomography (CT) scans. Unlike OCT, these imaging modalities have different structural characteristics, resolution requirements, and noise patterns. Directly applying PVGCN may not effectively capture relevant features, potentially impacting its performance. Future research could explore ways to adapt the model for these imaging techniques by modifying the graph construction strategy, integrating multi-modal information, or employing transfer learning to improve cross-modality generalization. While empirical studies have demonstrated that incorporating components like FFNs can mitigate over-smoothing in GNNs by enhancing feature expressivity and preserving feature diversity [35,36], a formal theoretical proof of this mechanism remains an open research question, that can be explored in our future works.

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